

Literature Review

SL No.	Paper Title	Applied on FACED?	Year	Dataset Information	Input Features	Graph Construction	Method / Purpose	Evaluation Strategy	Key Results	Research Gap
1	EEG Emotion Recognition Using Dynamical Graph Convolutional Neural Networks	No	2020	SEED : 15 subjects, 62 channels, 3 classes; DREAMER : 23 subjects, 14 channels, binary V/A/D	SEED : DE, PSD, DASM, RASM, DCAU; DREAMER : PSD	Channels as nodes; shared trainable adjacency matrix	DGCNN learns functional channel relationships for emotion classification	Subject-dependent and LOSO ; DREAMER session-independent	SEED : 90.40% subject-dependent, 79.95% LOSO; DREAMER : 86.23% V, 84.54% A, 85.02% D	A shared graph may miss subject/trial-specific connectivity ; limited temporal modelling
2	EEG-Based Emotion Recognition Using Regularized Graph Neural Networks	No	2022	SEED : 15 subjects, 62 channels, 3 classes; SEED-IV : 15 subjects, 62 channels, 4 classes	Multi-band differential entropy	Distance-based local edges, hemispheric edges, and learnable sparse symmetric adjacency	RGNN models biological connectivity; NodeDAT supports cross-subject adaptation; EmotionDL handles noisy labels	Subject-dependent and subject-independent experiments with ablation	Outperformed most reported baselines; informative prefrontal, parietal, occipital and hemispheric links identified	Uses some manual biological priors ; temporal dynamics remain limited
3	Cross-Subject EEG Emotion Recognition With Self-Organized Graph Neural Network	No	2021	SEED and SEED-IV : 15 subjects, 62 channels	Five-band DE with temporal segments	Input-specific adjacency via Softmax weighting and top-k sparsification	SOGNN combines temporal convolution, dynamic graph learning and GCN classification	LOSO cross-validation	86.81% on SEED; 75.27% on SEED-IV	Small datasets; dynamic graph adds complexity; limited neurological constraints
4	A Large Finer-Grained Affective Computing EEG Dataset	Yes	2023	FACED : 123 subjects, 32 channels, 28 videos, 9 emotions	Raw/preprocessed EEG; five-band DE and PSD	No graph model proposed	Introduces FACED and baseline intra-/cross-subject emotion recognition	Classical ML and contrastive-learning baselines	Nine fine-grained emotions were classifiable; larger subject pool than older datasets	Dataset paper only; FACED-9 cross-subject recognition remains difficult
5	LGGNet: Learning From Local-Global-Graph Representations for Brain-Computer Interface	No	2023	Three BCI datasets; DEAP : 32 subjects, 32 channels	Multi-scale temporal convolutions and power features	Predefined local brain-region graphs plus learned global similarity graph	LGGNet fuses temporal, local and global graph representations	Nested CV, accuracy/F1, ablation and visualization	DEAP emotion : 64.51% F1; preference: 63.07% accuracy, 72.53% F1	Local regions are manually defined; not designed for FACED-9
6	GNN4EEG: A Benchmark and Toolkit for EEG Classification With GNN	Yes	2024	FACED : 123 subjects, 32 channels, 28 videos, 9 emotions	Five-band DE with LDS smoothing	Channels as nodes; benchmarks DGCNN, RGNN, SparseDGCNN and HetEmotionNet	Standardized toolkit for comparing EEG-GNN models	Intra-/cross-subject CV, FCV and NCV protocols	Cross-9 : RGNN 31.3%; Intra-9 : HetEmotionNet 44.9% under NCV	Benchmark rather than a new model; FACED-9 generalization remains weak
7	EEG-Based Emotion Recognition Using GCN With Dual Attention Mechanism	No	2024	DEAP : 32 subjects, 32 channels; SEED/SEED-IV : 15 subjects, 62 channels	Four- or five-band DE	Fixed graph from 3D electrode distances	DAMGCN combines residual GCN with channel and frequency attention	10-fold CV, leave-one-trial-out, LOSO and ablation	Subject-dependent : 99.42% SEED, 96.86% SEED-IV; leave-one-trial-out : 78.21% DEAP, 90.40% SEED, 83.55% SEED-IV	A fixed distance graph may not capture trial-specific functional connectivity
8	Dynamic-Attention-Based EEG State Transition Modeling for Emotion Recognition	Yes	2025	FACED : 123 subjects, 32 channels, 28 videos, 9 emotions	Temporal and spatial transition representations	No explicit graph ; dynamic spatiotemporal state components	DAEST models EEG state transitions using dynamic attention and contrastive learning	Cross-subject evaluation	FACED-2 : 75.4%; FACED-9 : 59.3%; SEED: 88.1%; SEED-V: 73.6%	Not GNN-based; state components may be harder to interpret neurologically
9	Cross-Subject EEG Signals-Based Emotion Recognition Using Contrastive Learning	Yes	2025	FACED : 123 subjects, 32 channels, 9 emotions; also SEED, CEED and MPED	Spatial, temporal and frequency-domain features	No graph ; contrastive and domain-adaptation framework	CSCL learns subject-invariant representations with contrastive learning and DANN	Cross-subject and LOSO-style evaluation	FACED : 65.98% accuracy	Not GNN-based; comparisons depend on preprocessing and split protocols
10	EmT: A Novel Transformer for Generalized Cross-Subject EEG Emotion Recognition	Yes	2025	FACED : 123 subjects, 32 channels, 9 classes; MAHNOB-HCI for regression	Seven-band relative PSD from long/short segments	Temporal graph sequences with multiple learnable adjacency matrices	EmT combines multi-view pyramid GCNs and a temporal Transformer	Generalized cross-subject classification and regression	SEED : 0.802 accuracy, 0.821 F1; FACED : 0.608 accuracy, 0.740 F1	Complex architecture; learned graphs may be difficult to interpret ; FACED 9 remains challenging
11	MSGM: A Multi-Scale Spatiotemporal Graph Mamba for EEG Emotion Recognition	Yes	2026	FACED : 123 subjects, 32 channels, 28 videos, 9 emotions	Seven-band relative PSD with multi-scale windows	Adaptive local/global graphs using predefined brain regions	MSGM combines multi-scale temporal learning, GCNs and Mamba	Subject-independent leave-n-subject-out CV	FACED : 63.17% accuracy, 76.01% F1	Uses predefined regions; stimulus confounding was not directly tested
12	Stimulus Identity Rather Than Emotion Drives EEG Classification on the FACED Dataset	Yes	2026	FACED : 123 subjects, 32 channels, 28 videos, 9 emotions	Official FACED DE features	No graph ; LinearSVC and CLISA analyses	Tests on whether FACED models learn emotion or video identity	Intra-/cross-subject, self-report and reduced-stimulus analyses	Strong accuracy remained without matching self-reports; one-video-per-emotion sometimes improved results	Supports the need for stimulus-independent and self-report-aware evaluation